

YURY BELY

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Software Engineer with 15+ years of experience in systems and cross-platform development (C++, Rust, Python). Designed and built desktop, mobile, and real-time applications across Windows, Linux, macOS, Android, and iOS. Strong background in performance optimization, networking, and low-level system architecture. Focused on performance-critical systems and cross-platform architecture. Applied LLM tools for automation and development workflows.

Work Experience

ScienceSoft — IT software development

Mar 2024 – Sep 2025

Rust Developer

- Built cross-platform desktop applications using Rust, Tauri, and Vue.js for Windows, macOS, and Linux in a team of 3.
- Reduced application memory footprint by 10% compared to Electron-based alternatives.
- Improved startup time by 2x through native integration and optimized resource loading.

Ispirer — Database migration software

Jan 2021 – Mar 2024

Senior Software Engineer (C++, Python, SQL)

- Contributed to development of Oracle-compatible functionality for an enterprise RDBMS in a team of 5, implementing support for PL/SQL-like syntax and behavior.
- Built an Oracle compatibility mode enabling seamless migration by switching connection strings without changes to business logic or SQL queries.
- Implemented and validated support for hundreds of SQL/PLSQL constructs, including procedures, functions, and triggers.
- Developed Python-based tooling and automated tests to validate database behavior and ensure Oracle compatibility.
- Worked extensively with SQL and PL/SQL for feature development, testing, and validation.
- Resolved compatibility edge cases between Oracle and target DB, ensuring consistent query execution behavior.
- Reduced migration effort from months to days for existing enterprise systems.
- Ensured full backward compatibility with native database mode.

ScienceSoft — Internet, Software

Jun 2017 – Jan 2021

Senior C++ Developer

- Collaborated in a 30+ engineer cross-functional team to build a secure cross-platform video conferencing application.
- Built and maintained a shared C++ core library used across 4 platforms (iOS, Android, Windows, macOS), reducing code duplication and ensuring consistent behavior.
- Introduced adaptive media management supporting real-time speaker detection in calls with 5+ participants.
- Implemented local storage handling thousands of messages and session records per user, improving data reliability.
- Strengthened end-to-end encryption by replacing WebRTC SSL with OpenSSL, improving security compliance and control over cryptographic stack.
- Resolved concurrency issues and reduced crash rate on mobile platforms by 25%.
- Created internal testing and support tools, improving SDK reliability.

ScienceSoft — Healthcare, Software

Apr 2016 – Jun 2017

Senior C++ Developer

- Built a medical software system for auditory hallucination treatment in a team of 5, with real-time VoIP and voice transformation.
- Designed system architecture and developed a QML-based UI.
- Prototyped a system to improve GPS positioning accuracy using Real-Time Kinematic (RTK) and Continuously Operating Reference Stations (CORS) techniques.

ScienceSoft — Mobile development

Feb 2015 – Apr 2016

Xamarin Developer, .NET Developer

- Built mobile applications in teams of 3–7, integrating GPS, camera, and push notifications.
- Published applications to App Store and Google Play.
- Created an interactive 3D exhibition preview application in Unity.

ScienceSoft — Internet, Music, Software

Jan 2014 – Feb 2015

Senior C++ Developer

- Built a desktop music streaming application in a team of 3, based on WebKit with deep OS-level integration.

- Architected and built a real-time desktop and webcam streaming application.
- Implemented video capture and frame processing using OpenCV.
- Added audio streaming and mixing via GStreamer/FMOD.
- Integrated RTSP client for real-time video transmission.
- Delivered a Windows-based solution using C++, Qt, and STL.

ScienceSoft — Internet, Software

Aug 2012 – Jan 2014

Senior C++ Developer

- Ported a feature-rich Chromium-based browser to Windows in a team of 5, adding native support for torrent handling without the need for third-party applications.
- Contributed to interface localization and platform-specific adaptations.
- Built a video search and download manager with support for multiple formats and integration with leading video platforms (YouTube, Vimeo, Dailymotion, etc.), featuring an embedded video player.
- Built a feature-rich download manager with BitTorrent support and plugin architecture.

ScienceSoft — Internet, Software

Jun 2011 – Aug 2012

C++ Developer

- Integrated and maintained social network functionality via the Facebook API in a team of 5.
- Developed JavaScript solutions for interacting with Facebook services, improving automation and efficiency.

Science Research Institute of means of automation

Sep 2007 – Jun 2011

Junior C++ Developer

- Built a cross-platform application using Qt, implementing a complex UI and highly customized, non-standard design solutions.

Education

Belarusian State University of Informatics and Radioelectronics

2002 – 2007

Bachelor of Computer Science

Languages: English (B2)

Technical Skills

Core Expertise: Cross-platform architecture, concurrent and multithreaded programming, design patterns (SOLID, GoF), REST API integration, performance profiling and optimization

Languages: C++, Rust, Python, C, C#, PL/SQL, SQL, Swift, JavaScript/TypeScript

Frameworks and Libraries: Qt, QML, STL, BOOST, OpenSSL, Tauri, Djinni, OpenCV, WebRTC, GStreamer, Xamarin.Forms, Flex/Bison, OpenGL

Databases: SQLite, PostgreSQL, Oracle

Platforms: Windows, Linux, macOS, Android, iOS

Tools: Git, Docker, CMake, CI/CD, Visual Studio, Xcode, Qt Creator

AI/ML: Claude Code, GitHub Copilot, ChatGPT, local LLMs (Ollama), LLM API integration, prompt engineering, AI-assisted development